

Question Paper

Exam Date & Time: 20-Nov-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
SEVENTH SEMESTER B.TECH. DEGREE EXAMINATIONS - NOVEMBER 2025

Quantum Computing Practice and Application [CSE_40B4]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Examine the difference between parametrized and fixed quantum circuits, and discuss how Qiskit models, simulates, and mitigates noise when executing quantum algorithms. (5)
- A) Course Outcome: "CO1" / Cognitive Level: "Analysis"
- B) Conclude that how Qiskit connects to IBM Quantum cloud devices, including the role of accounts, providers, and backends in this process. Describe how this connection enables real hardware execution of quantum circuits. (5)
- Course Outcome: "CO1" / Cognitive Level: "Analysis"
- 2) Examine how quantum algorithms can potentially improve the efficiency or accuracy of K-Means, K-Median, and Hierarchical clustering methods. Compare the classical and quantum approaches in terms of (a) data representation, (b) distance computation, and (c) optimization process. (5)
- A) Course Outcome: "CO2" / Cognitive Level: "Application"
- B) Discover how the expressibility and trainability of a variational quantum circuit affect its performance. Discuss how issues like barren plateaus, entanglement structure, and ansatz design influence the convergence and generalization of VQMs in practical quantum machine learning applications. (5)
- Course Outcome: "CO2" / Cognitive Level: "Analysis"
- 3) Compare & Contrast Ground State Solvers and Excited State Solvers in Qiskit Nature. Why is finding the ground state energy an essential first step in molecular simulations? (5)
- A) Course Outcome: "CO3" / Cognitive Level: "Analysis"
- B) Examine how does the optimizer influence the accuracy and efficiency of Minimum Eigen solvers like VQE? Discuss common optimization challenges such as barren plateaus and local minima. (5)
- Course Outcome: "CO3" / Cognitive Level: "Application"
- 4) Justify the significance of Nash Equilibrium in the context of Quantum Game Theory, also explain how can quantum entanglement lead to equilibrium shifts or cooperative outcomes in non-cooperative games? (5)
- A) Course Outcome: "CO4" / Cognitive Level: "Analysis"

- B) Discover the possible role of quantum annealing in improving the security and scalability of blockchain networks. How can quantum-based optimization contribute to better consensus protocols? (5)

Course Outcome: "CO4" / Cognitive Level: "Evaluate"

- 5) Explain the different methods of encoding classical data into quantum states, such as amplitude encoding, basis encoding, and angle encoding. Compare their advantages and challenges. (5)

A)

Course Outcome: "CO3" / Cognitive Level: "Evaluate"

- B) Evaluate the potential of quantum-native data processing in future applications such as drug discovery, finance, and optimization. What unique advantages does quantum data offer in these domains? (5)

Course Outcome: "CO4" / Cognitive Level: "Evaluate"

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Event Name	Subject
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058-END SEMSTER-ODD-2025 CSE_40B4:Quantum Computing Practice and Application

#	Course Outcome	Description	Marks
1	1	CO 1	10
2	2	CO 2	10
3	3	CO 3	15
4	4	CO 4	15
Total			50

* **NA** - CO Not Selected

Event Name	Subject
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058-END SEMSTER-ODD-2025 CSE_40B4:Quantum Computing Practice and Application

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	0	0(0%)
3	Application	2	10(20%)
4	Analysis	5	25(50%)
5	Create	3	15(30%)
6	Evaluate	0	0(0%)
Total		10	50 (100%)